# CE 310 — Week 6 Assignment

## Empirical Percentiles Across CE Practice

Total: 60 points · Due: Wednesday of Week 7, 9:00 AM · Submit via D2L

Do not repeat thresholds already used in the lab (4,000 psi, 65 dB, 65 mph, 85th-percentile speed, $250/CY, 90th-percentile cost, the AM/PM 85th-percentile comparisons). This assignment uses new thresholds to extend, not duplicate, what you computed in class.

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## How to Submit

> ☐ Following this submission format exactly is essential — with a class this size, grading depends on it. Submissions that don't comply will be penalized.

| Requirement | Detail |
|---|---|
| File format | .xlsx only — written responses go in the Answer Sheet cells, not in a separate PDF |
| Sheet name | Must be exactly Assignment — do not rename or delete it |
| Column layout | Enter every answer in Column B — do not insert, delete, or move columns |
| Written responses | Type them in the Column B cells for P2b, P2c and P3 on the same sheet — do not submit a separate PDF |
| File name | CE310_W6_Assignment_<NetID>.xlsx (e.g. CE310_W6_Assignment_abc123.xlsx) |
| Submission | Upload the .xlsx file to D2L Dropbox before the deadline |

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## Problem 1 — Concrete Strength and Site Acoustics: New Thresholds (30 pts)

P1a–c use column D (fc_psi) of Week5_CE_Measurements.csv, all 500 records unless noted. P1d uses column C (NoiseLevel_dB) of Week6_Noise_Measurements.csv.

P1a (8 pts). What percentage of tests exceed 5,200 psi? Week 5's Normal model estimated this at 4.1% — report your empirical value and, in 1–2 sentences, compare.

P1b (8 pts). What percentage of tests fall strictly between 4,500 and 5,000 psi? Week 5's Normal model estimated this at 33.9% — report your empirical value and, in 1–2 sentences, compare. (This gap is larger than most you saw in lecture — say why you think that might be, in general terms.)

P1c (7 pts). Compute the empirical 5th percentile strength (the Eurocode 7 “characteristic value” convention — stricter than ACI 318’s 10th percentile). Week 5’s Normal model estimated this at 3,769 psi. Report your empirical value (exact-rank SMALL or interpolated PERCENTILE, your choice — state which) and compare to the model.

P1d (7 pts). What percentage of noise readings exceed 70 dB — a stricter threshold than the lab’s 65 dB HUD “Acceptable” line? Report your empirical value and, in 1–2 sentences, state what a site-acceptability review would conclude at this stricter threshold versus the lab’s 65 dB result (18.0%).

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## Problem 2 — Traffic Speed: New Thresholds and a New Percentile (18 pts)

P2a (6 pts, 2 pts each). What percentage of records show speed above 55 mph — (i) all records, (ii) AM Peak only, (iii) PM Peak only?

P2b — Written (6 pts). Report the 90th-percentile speed — a stricter design target than the lab’s 85th percentile — for (i) all records, (ii) AM Peak, (iii) PM Peak. In 3–5 sentences, explain what moving from the 85th to the 90th percentile does to a posted speed limit, and why a transportation agency might choose one over the other (connect back to the Part 1 debate from lecture).

P2c — Written (6 pts). A colleague argues: “Since Week 5’s Normal model and this week’s empirical values are close for most of these speed questions, we should just always use the Normal model — it’s faster.” In 3–5 sentences, argue for or against this position, citing at least one specific number from Problem 1 or 2 where the empirical and model-based answers diverged meaningfully.

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## Problem 3 — Reflection (12 pts)

Write a 100–150 word reflection citing at least 2 specific numbers from this assignment or the lecture. Prompt: A cost estimator on your team says, “AACE guidance is just a suggestion — I always budget to the mean, plus a flat 15% for contingency.” Using what you know about the empirical 90th-percentile cost ($287.56/CY, or $274.33/CY for Alpha-only, $288.86/CY for Beta-only, vs. a mean of $194.75/CY), argue for or against your colleague’s approach.

> On conventions: $287.56/CY is your lab B10 answer (exact rank, LARGE); your lab B11 answer, $286.50/CY, is the same percentile computed by interpolation (PERCENTILE). The supplier-specific figures above are interpolated. The two conventions differ by about a dollar here, which is far too small to change the argument — cite either, but say which you used.

## Grading Summary

| Problem | Points |
|---|---|
| P1 (a+b+c+d) | 30 |
| P2 (a + written b + written c) | 18 |
| P3 | 12 |
| Total | 60 |